

Examiner
only

3. Carbon-12 ($^{12}_6\text{C}$) and carbon-14 ($^{14}_6\text{C}$) are isotopes.

- (a) Explain, in terms of the **numbers of particles**, why carbon-14 has an unstable nucleus and carbon-12 has a stable nucleus. [3]

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- (b) An unstable nucleus becomes more stable following the emission of radiation. **Complete the table below.** [4]

Type of radiation	Symbol	Description
gamma		Electromagnetic (em) wave
alpha	$^4_2\alpha$	
beta		

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